MEMORANDUM

DATE: February 19, 2026
TO: Prof. Yanfei Xu
FROM: Group members: Tenzin Sharchung, Tim Ryan, Ben Schaffer
SUBJECT: Rockwell hardness tester (Mitutoyo HR-320MS) and Nanoindentation Tester (Anton Paar Hit 300) on two samples of W-1 Steel.

Purpose
Determine whether there is a statistically significant difference in the hardness of two samples of tempered W-1 “drill rod” steel using an unpaired students T Test. Conduct nanoindentation on samples to assess differences in Vickers hardness, plastic work, and elastic indentation modulus.

Results and Discussion
Rockwell hardness testing measures how deeply a standardized indenter penetrates a material under a specific load; this provides a numerical indicator of resistance to deformation. According to the null hypothesis, the W-1 tool Steel specimen tempered at 250 °C (Type B) is expected to have slightly lower hardness than that at 200 °C (Type A) due to more extensive tempering. Upon initial inspection, the Rockwell hardness testing results do in fact show higher hardness with Type A steel than Type B steel for both group and section data (Table 1). For the group data, Type A had a mean hardness of 61.68 HRC and Type B had a mean hardness of 58.98 HRC. For the section data, Type A had a mean hardness of 62.23 HRC and Type B had a mean hardness of 58.54 HRC. Both the subdataset and the complete section dataset had a difference of about 3-4 Rockwell units across the steels showing general agreement. The standard errors are relatively small against the difference in mean values, indicating sufficient measurement consistency. Statistically speaking, an unpaired two sample t-test assuming unequal variances conducted to determine whether the difference in mean hardness values was significant. The calculated t-statistic $t = 13.10$ with 50 degrees of freedom far exceeded the critical t value at the 95% confidence level of $t_{crit} = 2.01$. In addition, the corresponding p-value was confirmed to be far less than 0.05 using both Excel and MATLAB based calculations indicating a significant statistical difference in the hardness of the two steel types. Therefore tempering temperature had a significant influence on mean hardness. The 95% confidence interval for the difference in mean hardness was 3.02 to 4.12 HRC, indicating that Type A steel is consistently harder than Type B steel.

To further validate findings, nanoindentation was performed using a nanoindentor. Unlike Rockwell testing, nanoindentation continuously records load and displacement during indentation allowing for extraction of both hardness and elastic indentation modulus. In addition, plastic work can be gathered. The group used a nanoindentation load of 100 mN. The nanoindentation results confirmed that the specimen tempered at 200 °C (Type A) exhibited higher hardness and greater resistance to plastic deformation compared to the 250 °C specimen (Type B), reinforcing the conclusions drawn from Rockwell testing.

The variation in data between group and section is attributed to an inaccuracy in the Rockwell Hardness Tester used by group A7. When calibrating the machine with a material with known HRC 61.8 +/- 0.5, the results were an average HRC of 61.16, which lies out of tolerance. The machine was known to provide inaccurate yet precise results by lab staff, leading to the group providing a lower mean value for HRC.

Table 1: Mean, Standard Deviation, and Standard Error for Hardness of Type A and B Steel

	Type A			Type B		
	$\bar{\chi}$	S_x	$\varepsilon_{\bar{x}}$	$\bar{\chi}$	S_x	$\varepsilon_{\bar{x}}$
Group Data	61.68	0.38	0.19	58.98	0.51	0.26
Section Data	62.23	0.65	0.16	58.54	1.09	0.28

Table 2: Statistical Analysis Equations

Quantity	**Equation**
Sample Mean	$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$
Sample Standard Deviation (MATLAB std)	$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$
Sample Size	$n =$ number of observations
Degrees of Freedom (Single Sample)	$df = n - 1$
Standard Error of Mean	$SE = \frac{s}{\sqrt{n}}$
Student t Critical Value	$t_{\alpha/2,df} = T^{-1}(1 - \alpha/2, df)$
Confidence Interval	$CI = \bar{x} \pm t_{\alpha/2,df}\left(\frac{s}{\sqrt{n}}\right)$
Unpaired Two-Sample t Statistic	$t = \frac{\bar{x}_A - \bar{x}_B}{\sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}}$
Welch Degrees of Freedom (MATLAB ttest2)	$df = \frac{\left(\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}\right)^2}{\frac{\left(\frac{s_A^2}{n_A}\right)^2}{n_A - 1} + \frac{\left(\frac{s_B^2}{n_B}\right)^2}{n_B - 1}}$

$t_{\alpha,v}$ Manual Computation Steps:
For 95% confidence (2 tailed) independent samples T-test use $\alpha/2 = 0.05/2 = 0.025$
Using the above formula for degrees of freedom with $s^2A = 0.60$ and $S^2B = 1.78 \rightarrow$ dof = 50 (approx.)
From the table, $t_{0.025,\ 50}$ returns a critical value of 2.009 (Result Confirmed with Excel Based Calculations)

Table 3: Unpaired t-test Results Excel

Mean A	Mean B	T statistic	DOF	p-value	CI Lower	CI Upper
62.011875	58.44	13.0961328	49.76409686	9.55775E-18	3.023934857	4.119815143

Table 4:Unpaired t-test Results Matlab

Mean A	Mean B	t statistic	DOF	p-value	CI Lower	CI Upper
61.9856	58.5106	14.1696	70	3.0038e-22	2.9859	3.9641
Result: Reject Null Hypothesis						

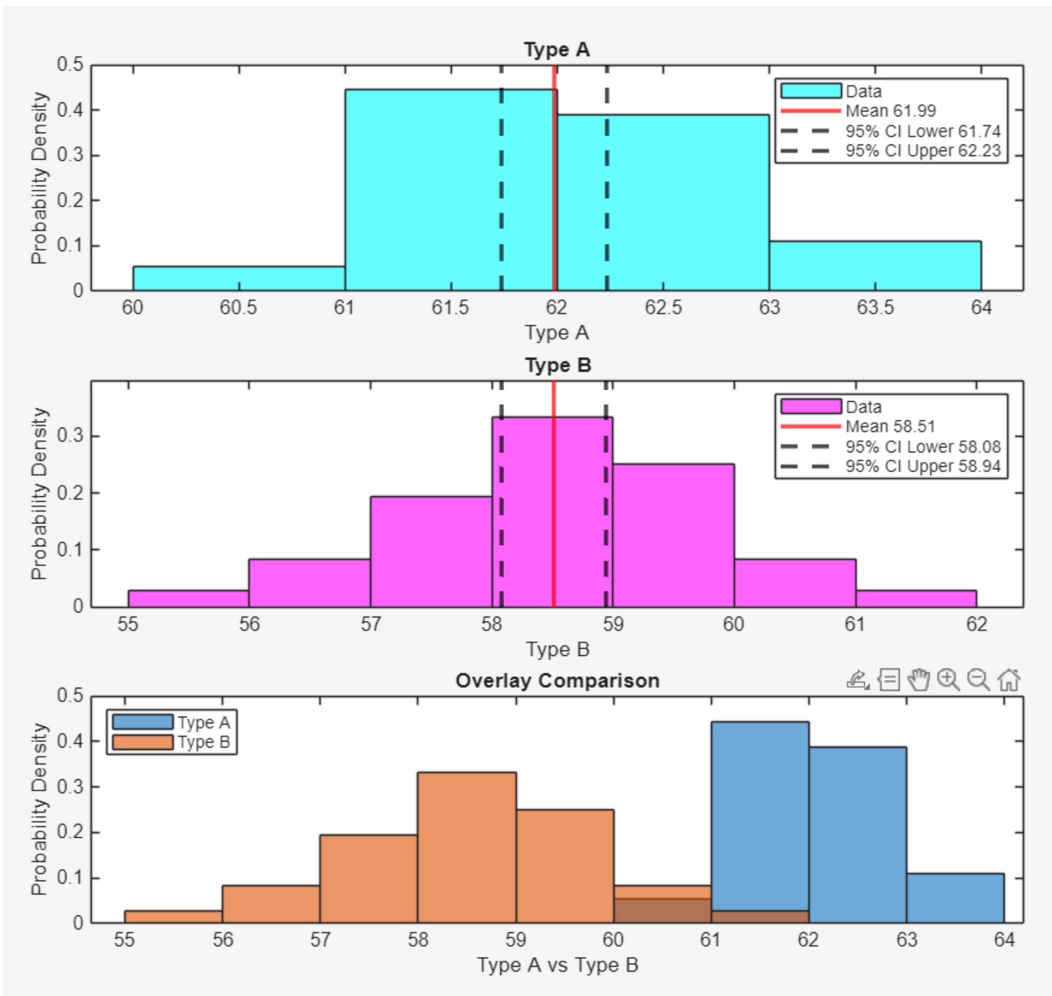

Figure 1: Histogram of Type A hardness (top), Histogram of Type B hardness (middle), Histogram of Type A and B hardness overlaid (bottom)

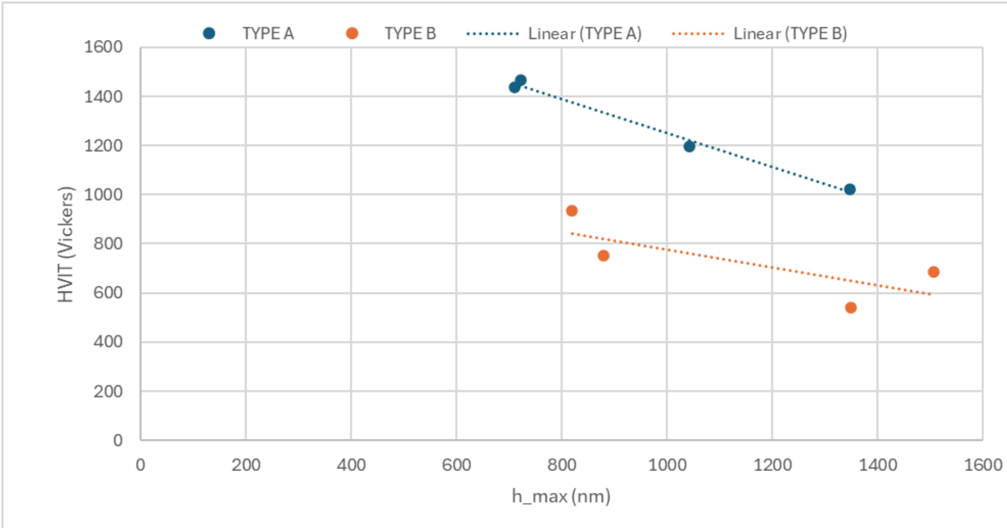

Figure 2: Scatterplot of HVIT (Vickers hardness) values for Type A and Type B as a function of maximum displacement, h_{max}, from Nanoindentation tester data.

Nanoindentation tester data shows results consistent with increased hardness associated with Type A steel samples over Type B steel. Plots from the test use data from 4 tests of each steel type with varying max loads. Figure 2 shows that HVIT decreases with increasing maximum displacement for either Type A or Type B steel. This reflects a lower measured hardness with a larger load and indentation depth. Type A shows consistent higher hardness values, indicating that the lower tempering temperature produced a harder steel microstructure.

In Figure 3, at comparable indentation depths, Type B generally exhibits similar or slightly higher plastic work than Type A samples. This is consistent with the lesser hardness of Type B steel samples. Greater plastic work is likely observed as softer materials undergo greater plastic deformation.

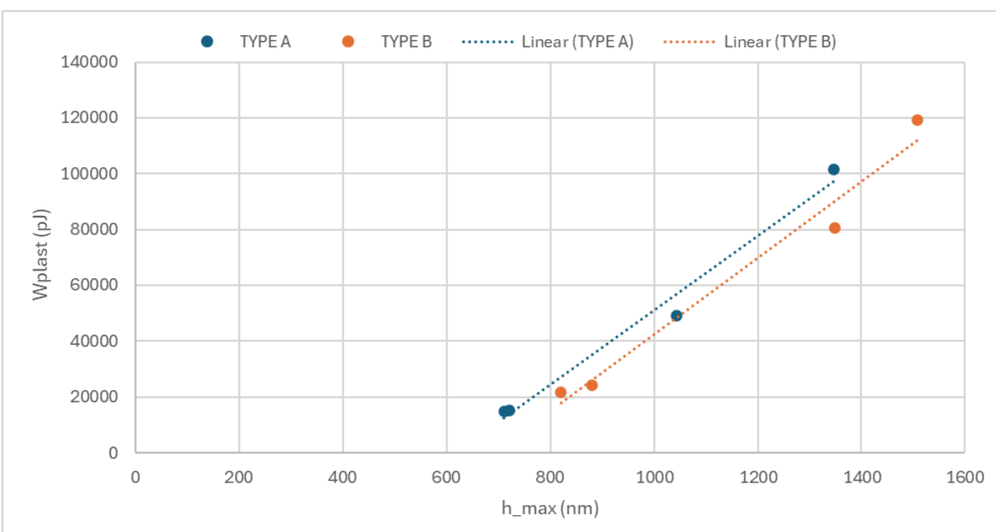

Figure 3: Scatterplot of plastic work or Wplast values for Type A and Type B as a function of maximum displacement, h_{max}, from Nanoindentation tester data.

Results also show that the elastic indentation modulus (EIT) is not significantly varied between Type A and Type B steel samples. From Figure 4, Type A steel shows a slight decreasing EIT value trend and Type B shows a slight increasing EIT value trend. The range of EIT values for both samples is roughly the same and overall variation is small.

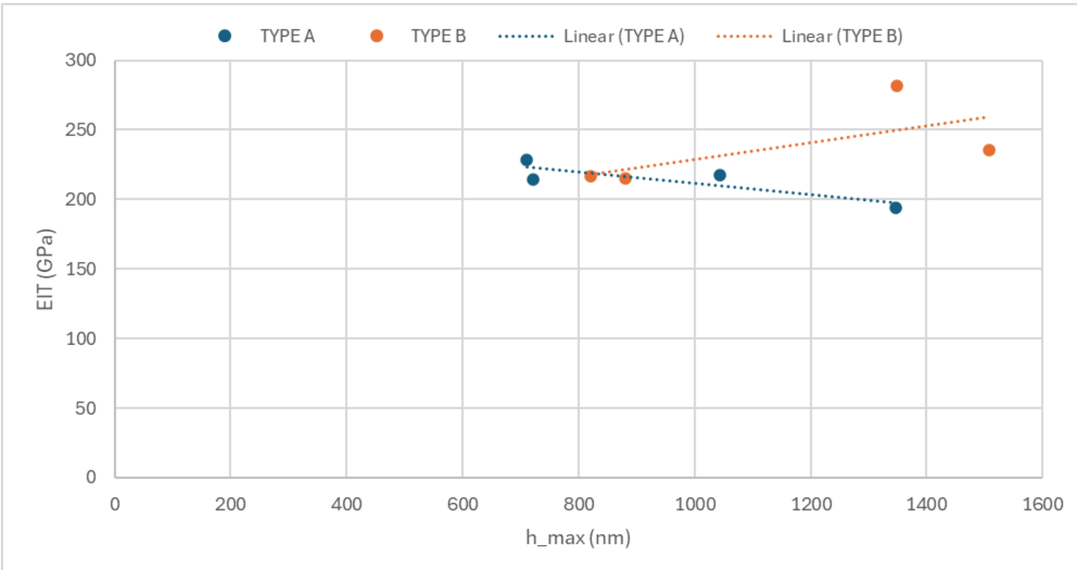

Figure 4: Scatterplot of elastic indentation modulus or EIT values for Type A and Type B as a function of maximum displacement, h_{max}, from Nanoindentation tester data.

Conclusion

At a 95% confidence level ($\alpha = 0.05$) both the p-value of 3.0e-22 obtained using MATLAB and the p-value of 9.56e-18 obtained using Excel require rejection of the null hypothesis. Furthermore the t-statistics of 13.1 and 14.2 calculated using Excel and MATLAB respectively are far higher than the critical table value of 2.009. These results indicate a statistically significant difference between the two groups being compared. As anticipated, Type A demonstrated higher average hardness than Type B.

Nanoindentation testing further confirmed differences in hardness and plastic deformation behavior. Type A exhibited higher hardness and slightly lower plastic work, consistent with a harder martensitic microstructure resulting from tempering at a lower temperature. In contrast, Type B steel showed reduced hardness and greater plastic work, indicating increased plastic deformation resulting from tempering at a higher temperature. Nanoindentation results also showed elastic indentation modulus did not vary significantly between samples, suggesting that tempering has minimal influence on elastic stiffness.

Some variation in hardness measurements was attributed to calibration inaccuracies in the Rockwell hardness tester used during group testing. Future experiments could be improved through a better calibration process, ensuring that the machines provide an accurate measurement of hardness prior to testing.

Contributions

Table 5: Group Member Contributions

Member	**Key Contributions**
Tenzin Sharchung	- Experimental Testing (Rockwell & Nanoindentation) - Data Collection - Nanoindentation Tester Data Plots - Nanoindentation Results and Conclusion Comments
Tim Ryan	- Experimental Testing (Rockwell & Nanoindentation) - Data Collection - Programmed unpaired t-test, plots, in Matlab - Created group and section data table. - Results, Discussion, and Conclusion reporting
Ben Schaffer	- Experimental Testing (Rockwell & Nanoindentation) - Data Collection - Results, Discussion & Conclusion Reporting - Computation of t_{crit} from table - Excel-based Calculations and hypothesis testing

References

[1] MathWorks, “ttest2,” MathWorks Documentation, https://www.mathworks.com/help/releases/R2025b/stats/ttest2.html, Accessed: March 1, 2026.